MATH1120 - Partial Differential Equations

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Introduction

> In order to solve this differential equation you look at it until a solution occurs to you.
>
> George Polya

title

lorem ipsum

- author

Chapter 1

Where PDEs Come From

1.1 Types of Partial Differential Equations

Partial differential equations are equations involving partial derivatives and several variables. We typically denote partial derivatives of u

$$u_x \equiv \frac{\partial u}{\partial x}$$

A function u is said to be a solution of the PDE if it satisfies the equation for every x, y in the domain.

Specifically, a solution that satisfies the equation for every x, y is a classical solution, whereas one that does not is called a weak solution. We will not cover these in the course.

1.1.1 Order of a PDE

The order of a PDE is the order of the highest-order derivative (of *any* variable) that appears in the equation. Formally:

$$\begin{aligned} \text{1st order PDE} &\equiv F(x, y, u(x, y), u_x, u_y) \\ \text{2nd order PDE} &\equiv F(x, y, u, u_x, u_y, u_{xx}, u_{yy}, u_{xy}) \end{aligned}$$

Example $u_x + u_y = 0$ is a first order PDE, $u_{xy} + u$ is a second order PDE.

1.1.2 Linearity of a PDE

We conventionally consider the left side of the equation an **operator** applied to the function in question. For example,

$$u_x + 2u_y = 0 \Rightarrow (\frac{\partial}{\partial x} + 2\frac{\partial}{\partial y})u = 0$$

The operator (and the equation in question) is called **linear** if it satisfies

$$\begin{cases} \mathcal{L}(u + v) = \mathcal{L}u + \mathcal{L}v \\ \mathcal{L}(cu) = c\mathcal{L}u \end{cases}$$

We can check whether an operator is linear by substituting $u+v$ and cu into u and checking whether the result is linear.

EXAMPLE $u_t - u_{xxt} + uu_x = 0$ is not a linear PDE:

$$\begin{aligned}
\mathcal{L} &= \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x^2 \partial t} + u\frac{\partial}{\partial x}\right) \\
\mathcal{L}(u+v) &= (u+v)_t - (u+v)_{xxt} + (u+v)(u+v)_x \\
&= u_t + v_t - u_{xxt} - v_{xxt} + (u+v)(u_x + v_x) \\
&= u_t + v_t - u_{xxt} - v_{xxt} + uu_x + uv_x + vu_x + vv_x \\
&\neq \mathcal{L}(u) + \mathcal{L}(v)
\end{aligned}$$

1.2 Homogeneity

If the sum of all instances of u and its derivative is zero, the equation is considered homogeneous. If this is not the case, the equation is inhomogeneous, and the leftover term is called the inhomogeneity.

1.3 Inhomogeneous Equations and Linearity

Suppose an inhomogeneous equation with a linear differential operator $\mathcal{L}$

$$\mathcal{L}u = h$$

Suppose we know one solution of it, f. Then, by linearity, the general solution is some

$$u = f + g$$

Where g is the general solution of the homogeneous equation $\mathcal{L}u = 0$.

1.4 1st Order Linear Constant Coefficient Equation

First order linear partial differential equations of constant coefficients are of the form

$$\boxed{au_x + bu_y = 0}$$

1.4.1 Geometric Method

We can separate the differential operator:

$$\left(a\frac{\partial}{\partial x} + b\frac{\partial}{\partial y}\right)u = 0$$

Recall that the directional derivative of a function is $\nabla u \cdot \hat{\mathbf{n}}$. Then the above implies that the directional derivative of u along the vector (a, b) is zero, and that the function is constant in that direction. Specifically, it will be constant along any line parallel to (a, b). These are the lines

$$y = \frac{b}{a}x + c \tag{1.1}$$

for an arbitrary c. These are called **characteristic lines**. Then u depends solely on C, and we can write

$$u = u(c) = f(y - (b/a)x)$$

where we just solved for c using 1.1, and where f is an arbitrary function. Then, if we are given a condition for, say, $x = 0$, we can just set that equal to f, and plug in $y - \frac{b}{a}x$.

Example Solve $2u_x - 3u_y = 0$. The operator is

$$(2\frac{\partial}{\partial x} - 3\frac{\partial}{\partial y})u = 0$$

The direction is $(2, -3)$, and the characteristic line is $y = -\frac{3}{2}x + c$. Then u is some

$$u = f(c) = f(y + \frac{3}{2}x)$$

We are given $u(0, y) = y^2$. Then plug in $y + \frac{3}{2}x$, and we get

$$u = f(y + \frac{3}{2}x) = \left(y + \frac{3}{2}x\right)^2$$

1.4.2 Change of Variables/Coordinate Method

We are still trying to solve

$$au_x + bu_y = 0$$

The idea is to rotate the coordinate system so that solution only depends on some x'. We perform a linear transformation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ b & -a \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Giving us

$$\begin{cases} x' = ax + by \\ x' = bx - ay \end{cases}$$

Then we use these to substitute u_x, u_y via the chain rule:

$$\begin{aligned} u_x = \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial y} \\ &= u_{x'} \cdot a + u_{y'} \cdot b \\ u_y = u_{x'} \cdot \frac{\partial x'}{\partial y} + u_{y'} \cdot \frac{\partial y'}{\partial y} &= u_{x'} \cdot b + u_{y'} \cdot (-a) \end{aligned}$$

And we plug into the original equation

$$\begin{aligned} a(u_{x'} \cdot a + u_{y'} \cdot b) + b(u_{x'} \cdot b + u_{y'} \cdot (-a)) &= 0 \\ a^2 u_{x'} + abu_{y'} + b^2 u_{x'} - abu_{y'} &= 0 \end{aligned}$$

And we will, without fail, get cancellation that leads to an ODE:

$$(a^2 + b^2)u_{x'} = 0$$

Then $u = f(y')$, where f is arbitrary, and, substituting,

$$u = f(ax - by)$$

The advantage of the change of variables method is that it works for inhomogeneous equations as well, whereas it is much more difficult to use the geometric method in that case. Solve inhomogeneous equations by setting the left side of the equation in the proper form.

Example Solve $u_x + u_y = 2u$.

$$\begin{cases} x' = x + y \\ y' = x - y \end{cases}$$

$$\begin{aligned} u_x &= u_{x'} \frac{\partial x'}{\partial x} + u_{y'} \frac{\partial y'}{\partial x} = u_{x'} + u_{y'} \\ u_y &= u_{x'} \frac{\partial x'}{\partial y} + u_{y'} \frac{\partial y'}{\partial y} = u_{x'} - u_{y'} \end{aligned}$$

Substituting, this simplifies to

$$\begin{aligned} 2u_{x'} &= 2u \\ \frac{\partial u}{\partial x'} &= u \\ \frac{1}{u}\,\mathrm{d}u &= \mathrm{d}x' \\ \ln|u| &= x' + f(y') \\ u &= e^{f(y')}e^{x'} \end{aligned}$$

f is arbitrary so denote $e^{f(y')} = f(y')$, and

$$\begin{aligned} u &= f(y')e^{x'} \\ u &= f(x - y)e^{x+y} \end{aligned}$$

Chapter 2

Wave and Diffusion Equations

2.1 Initial and Boundary Conditions

Initial conditions are conditions imposed when $t = 0$. Boundary conditions are imposed when x is some specific value; usually, we are considering values between 0 and some length l, in which case boundary conditions are values at $x = 0, l$.

In higher dimensions, boundary value conditions occur at some boundary surface denoted ∂D, where $D \subseteq \mathbb{R}^n$, and where $\hat{\mathbf{n}}$ is the outer normal vector.

Initial conditions for the heat equation look like

$$\begin{cases} u_t = k\nabla u \\ u(x, t_0) = \phi(x) \end{cases}$$

and for the wave equation, we need *two* initial conditions because the highest time derivative is of order two:

$$\begin{cases} u_{tt} = k\nabla u \\ u(x, t_0) = \phi(x) \\ u_t(x, t_0) = \psi(x) \end{cases}$$

2.1.1 Types of Conditions

1. **Dirichlet condition:** u is specified on the boundary:

$$u(x, t) = f(x, t) \text{ for } x \in \partial D$$

2. **Neumann condition:** directional derivative $\frac{\partial u}{\partial \hat{\mathbf{n}}}$ is specified, where $\hat{\mathbf{n}}$ is the outer normal vector:

$$\frac{\partial u}{\partial \hat{\mathbf{n}}} = g(x, t) \text{ for } x \in \partial D$$

3. **Robin condition:** $\frac{\partial u}{\partial \hat{\mathbf{n}}} + au$ is specified (both u and its directional derivative.)

For example, with a fluid bounded by a container, the flux of substance in that fluid through the boundary of the container is zero, so, by Fick's law ($\mathbf{J}$ is the diffusion flux vector),

$$\begin{aligned} \mathbf{J} &= -k\nabla u \\ 0 &= -k\nabla u \text{ for } x, y, z, \in \partial D \end{aligned}$$

2.2 The Wave Equation

Consider a flexible, elastic, homogeneous string or thread of length l, which undergoes relatively small transverse vibration. Let $u(x,t)$ be its displacement from equilibrium position at time t and position x. Because the string is flexible, the tension T points tangentially along the string. Let $T(x,t)$ be the magnitude of this tension vector. Let ρ be the density of the string. We know it to be constant.

Write down Newton's law for the part of the string between any two points at $x = x_0$ and $x = x_1$. Slope of string at x_1 is $u_x(x_1, t)$. Newton's law in the longitudinal and transverse directions are

$$\begin{aligned}\left.\frac{T}{\sqrt{1+u_x^2}}\right|_{x_0}^{x_1} &= 0\\ \left.\frac{Tu_x}{\sqrt{1+u_x^2}}\right|_{x_0}^{x_1} &= \int_{x_0}^{x_1} \rho u_{tt}\,\mathrm{d}x\end{aligned}$$

Where the left side we obtained by constructing an, infinitesimal triangle for length of the string, with sides 1, u_x, and $\sqrt{1+u_x^2}$, then using $\cos\theta = \text{Adj/Hyp}$ and $\sin\theta = \text{Opp/Hyp}$ to solve for the component of T along the transverse and longitudinal directions. We assume no motion along longitudinal direction, so the right side of the first equation is zero. Then, we use Newton's law, for the right side of the bottom equation, with u_{tt} being acceleration.

Assume the motion is small, and that $|u_x|$ is small. Then replace $\sqrt{1+u_x^2}$ with one. (We can justify this by using the Taylor expansion $\sqrt{1+u_x^2} = 1 + \frac{1}{2}u_x^2 + \ldots$). Then

$$\begin{aligned}\left.\frac{T}{u_x}\right|_{x_0}^{x_1} &= \int_{x_0}^{x_1} \rho u_{tt}\,\mathrm{d}x\\ T(u_x)_x &= \rho u_{tt}\end{aligned}$$

Assuming T is independent of x and t,

$$\boxed{u_{tt} = c^2 u_{xx}}$$

Where $x = \sqrt{T/\rho}$.

2.3 Solving the Wave Equation

2.3.1 Change of Variables Method

Our equation is

$$u_{tt} = c^2 u_{xx} \Rightarrow u_{tt} - c^2 u_{xx} = 0$$

We start by "factoring" the differential operator:

$$\begin{aligned}\left(\frac{\partial^2}{\partial t^2} - c^2\frac{\partial^2}{\partial x^2}\right)u &= 0\\ \left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)u &= 0\end{aligned}$$

We use a similar procedure as with linear equations, performing a change of variables. We choose variables that are "perpendicular" to the directional derivatives given by the linear operators:

$$\begin{cases} \xi = ct + x \\ \eta = ct - x \end{cases}$$

Then, by the chain rule:

$$\begin{aligned} \frac{\partial}{\partial t} &= \frac{\partial}{\partial \xi}\frac{\partial \xi}{\partial t} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial t} = c\frac{\partial}{\partial \xi} + c\frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial x} &= \frac{\partial}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial x} = c\frac{\partial}{\partial \xi} - c\frac{\partial}{\partial \eta} \end{aligned}$$

Plugging in to the factored operators above:

$$\begin{aligned} \frac{\partial}{\partial t} - c\frac{\partial}{\partial x} &= \left(c\frac{\partial}{\partial \xi} + c\frac{\partial}{\partial \eta}\right) - c\left(\frac{\partial}{\partial}\right)\xi - \frac{\partial}{\partial \eta}) \\ &= 2c\frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial t} + c\frac{\partial}{\partial x} &= \left(c\frac{\partial}{\partial \xi} + c\frac{\partial}{\partial \eta}\right) + c\left(\frac{\partial}{\partial}\right)\xi - \frac{\partial}{\partial \eta}) \\ &= 2c\frac{\partial}{\partial \xi} \end{aligned}$$

Substituting into the original equation, we have

$$\begin{aligned} \left(2c\frac{\partial}{\partial \eta}\right)\left(2c\frac{\partial}{\partial \xi}\right)u &= 0 \\ \frac{\partial^2}{\partial \eta \partial \xi}u &= 0 \end{aligned}$$

Which we can then integrate one variable at a time:

$$\begin{aligned} (u_\xi)_\eta &= 0 \\ u_\xi &= f(\eta) \\ u &= \int f(\xi)\,\mathrm{d}\xi + g(\eta) \end{aligned}$$

$\int f(\xi)\,\mathrm{d}\xi$ is also some arbitrary function of ξ, so we abuse notation to

$$u = f(\xi) + g(\eta)$$

And substituting our original variables:

$$u(x,t) = f(ct + x) + g(ct - x)$$

2.3.2 General and Particular Solution Method

We once again return to

$$\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)u = 0$$

We choose to denote

$$v = \left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)u$$

and thus

$$\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)v = 0$$

We're left with a system of equations

$$\begin{cases} (\frac{\partial}{\partial t} - c\frac{\partial}{\partial x})v = 0 \\ (\frac{\partial}{\partial t} + c\frac{\partial}{\partial x})u = v \end{cases}$$

We know that $v(x,t) = h(x+ct)$, where h is arbitrary, as the upper equation is a homogeneous linear equation. Then we're left with the bottom equation:

$$u_t + cu_x = h(x+ct)$$

Which is an inhomogeneous equation. By section 1.3, we know that some solution to

$$u_t + c_x = 0$$

will be a component of the general solution. We will call this $g(x-ct)$.

Now we just need to find the particular solution, which should be of the form $f(x+ct)$. We let $s = x+ct$ for simplicity, and apply this linear operator to f in order to solve the equation:

$$\begin{aligned} \left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)f &= h(x+ct) \\ \frac{\partial}{\partial t}(f(x+ct)) + c\frac{\partial}{\partial x}(f(x+ct)) &= h(s) \\ cf'(x+ct) + cf'(x+ct) &= h(s) \\ 2cf'(x) &= h(s) \\ f'(s) &= \frac{h(s)}{2c} \\ f(s) &= \frac{1}{2c}\int h(s)\,\mathrm{d}s \end{aligned}$$

And so

$$u(x,t) = g(x-ct) + f(x+ct)$$

To summarize, we solved the topmost homogeneous equation for v, leading to some $h(x+ct)$. Then we set that solution h equal to the bottom-most inhomogeneous equation.

To solve that inhomogeneous equation, we solve for the general solution $\mathcal{L}u = 0$, which gave us some $g(x-ct)$, as per linear homogeneous equations.

Then we guessed a particular solution of form $f(x+ct)$ in order to set $\mathcal{L}f = h$ collapse the equation to an ordinary differential equation.

2.3.3 Generalized Change of Variables

So, why did we make the change of variables

$$\begin{cases} \xi = ct + x \\ \eta = ct - x \end{cases}$$

Consider the general case

$$\left(a_{11}\frac{\partial}{\partial t} - a_{12}\frac{\partial}{\partial x}\right)\left(a_{21}\frac{\partial}{\partial t} + a_{22}\frac{\partial}{\partial x}\right)$$

Then we assume the change of variables will be of a form

$$\begin{cases} \xi = b_{11}t + b_{12}x \\ \eta = b_{21}t + b_{22}x \end{cases}$$

And now we are left with the task of finding b_{ij}. We effectively have a linear transformation of the form

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

We abbreviate the matrix with entries b_{ij} B. Then notice that

$$\begin{aligned} \frac{\partial}{\partial t} &= \frac{\partial}{\partial \xi}\frac{\partial xi}{\partial t} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial t} = b_{11}\frac{\partial}{\partial \xi} + b_{21}\frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial x} &= \frac{\partial}{\partial \xi}\frac{\partial xi}{\partial x} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial x} - b_{12}\frac{\partial}{\partial \xi} + b_{22}\frac{\partial}{\partial \eta} \end{aligned}$$

And thus, in matrix form,

$$\begin{aligned} \begin{bmatrix} \frac{\partial}{\partial t} \\ \frac{\partial}{\partial x} \end{bmatrix} &= \begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{bmatrix} \\ &= B^T \begin{bmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{bmatrix} \end{aligned}$$

And this implies that

$$\begin{bmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{bmatrix} = \left(B^T\right)^{-1} \begin{bmatrix} \frac{\partial}{\partial t} \\ \frac{\partial}{\partial x} \end{bmatrix}$$

We want the new variables ξ, η to be parallel to the directional derivatives defined by a_{ij}. In symbols,

$$\frac{\partial}{\partial \xi} \parallel a_{11}\frac{\partial}{\partial t} + a_{12}\frac{\partial}{\partial x}$$
$$\frac{\partial}{\partial \eta} \parallel a_{21}\frac{\partial}{\partial t} + a_{22}\frac{\partial}{\partial x}$$

Then, in matrix form

$$\begin{bmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{bmatrix} = c \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Defining the matrix with entries $a_{ij} \equiv A$, and returning to B,

$$\begin{aligned} \left(B^T\right)^{-1} &= A \\ B^T &= A^{-1} \\ B &= \left(A^{-1}\right)^T \end{aligned}$$

Now we know the change of variables matrix is the transpose of the inverse of the matrix A. In symbols:

$$\boxed{\text{Change of variables matrix} = \left(A^{-1}\right)^T}$$

For differential operator $\mathcal{L} = \left(a_{11}\frac{\partial}{\partial t} - a_{12}\frac{\partial}{\partial x}\right)\left(a_{21}\frac{\partial}{\partial t} + a_{22}\frac{\partial}{\partial x}\right)$.

2.3.4 Generalized Particular Solution Method

We once again have the differential operation

$$\left(a\frac{\partial}{\partial t} - b\frac{\partial}{\partial x}\right)\left(c\frac{\partial}{\partial t} + d\frac{\partial}{\partial x}\right)u = 0$$

We denote

$$v = \left(a\frac{\partial}{\partial t} + b\frac{\partial}{\partial x}\right)u$$

Then v satisfies

$$\left(c\frac{\partial}{\partial t} - d\frac{\partial}{\partial x}\right)v = 0$$

And our system of equations is

$$\begin{cases} \left(c\frac{\partial}{\partial t} - d\frac{\partial}{\partial x}\right)v = 0 \\ \left(a\frac{\partial}{\partial t} + b\frac{\partial}{\partial x}\right)u = v \end{cases}$$

$v(x,t) = h(dx + ct)$, where h is arbitrary. Then second equation becomes

$$au_t + bu_x = h(dt + cx)$$

The general solution will be of form

$$g(bt - ax)$$

And we choose the particular solution to be some $f(dt + cx)$ such that

$$\begin{aligned} a\frac{\partial}{\partial t}f(dt+cx) + b\frac{\partial}{\partial x}f(dt+cx) &= h(s) \\ ad \cdot f'(dt+cx) + bc \cdot f'(dt+cx) &= h(s) \\ (ad+bc)f'(s) &= h(s) \\ f'(s) &= \frac{1}{ad+bc}h(s) \\ f(s) &= \frac{1}{ad+bc}\int h(s)\,\mathrm{d}s \end{aligned}$$

Then the solution is of the form

$$u(x,t) = g(bt - ax) + f(dt + cx)$$

2.4 Wave Equation - Initial Value Problem

$$\begin{cases} u_{tt} = c^2 u_{xx} \\ u(x,0) = \phi(x) \\ u_t(x,0) = \psi(x) \end{cases}$$

We know the solution is of the form $u = f(x+ct) + g(x-ct)$. Let us plug in $t = 0$,
$u = f(x) + g(x) = \phi(x)$
And let us find u_t then plug in $t = 0$:

$$\begin{aligned} cf'(x+ct) - cg'(x-ct) &= \psi(x) \\ cf'(x) - cg'(x) &= \psi(x) \end{aligned}$$

Now we leverage the fact that f and g just became single variable functions:

$$\begin{cases} f' + g' = \phi' \\ g' - g' = \frac{1}{c}\psi \end{cases}$$

Then, solving for f' and g':

$$\left\{ f' = \tfrac{1}{2}\big(\phi' + \tfrac{1}{c}\psi\big) g' = \tfrac{1}{2}\big(\phi' - \tfrac{1}{c}\psi\big) \right.$$

Then, recognizing that f and g are functions of some $x \pm ct$, and integrating,

$$\begin{cases} f(r) = \frac{1}{2}\psi(r) + \frac{1}{2c}\int \psi\,\mathrm{d}r + A \\ g(s) = \frac{1}{3}\psi(s) - \frac{1}{2c}\int \psi\,\mathrm{d}s + B \end{cases}$$

We know that $u(x,0) = f + g = \phi(x)$, so $A + B = 0$, and then

$$\begin{aligned} u(x,t) &= f(x+ct) + g(x-ct) \\ &= \frac{1}{2}\phi(x+ct) + \frac{1}{2c}\int_0^{x+ct} \psi(s)\,\mathrm{d}s + \frac{1}{2}\phi(x-ct) - \frac{1}{2c}\int_0^{x-ct} \psi(s)\,\mathrm{d}s \end{aligned}$$

By $\int_a^b f\,\mathrm{d}s = -\int_b^a f\,\mathrm{d}s$, we can combine the integrals and

$$\boxed{u(x,t) = \frac{1}{2}\phi(x+ct) + \frac{1}{2}\phi(x-ct) + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\,\mathrm{d}s}$$

What we get is actually quite intuitive: ϕ is a function that depends on x, but gets transposed to the right and to the left as time increases due to the $\pm ct$ term in the input. Then, we get a "propagation" of ϕ at a speed c.

The ψ term, since it's under an integral, effectively *expands* over x as t increases. We see that the limits of integration become larger as t goes increases, and so the area covered gets larger and we see an expanding graph.

2.5 Energy of Vibrating String

$$\frac{\mathrm{d}}{\mathrm{d}t}KE = \rho\int_{-\infty}^{\infty} u_t u_{tt}\,\mathrm{d}x$$

Then, by $\rho u_{xx} = T u_{xx}$,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}KE &= T\int_{\infty}^{\infty} u_t u_{xx}\,\mathrm{d}x \\ &= T\left(\int_{-\infty}^{\infty} (u_t u_x)_x\,\mathrm{d}x - \int_{-\infty}^{\infty} u_{tx}u_x\,\mathrm{d}x\right) \\ &= T\left((u_t u_x)\Big|_{-\infty}^{\infty} - \frac{1}{2}\int_{\infty}^{\infty} \frac{\partial}{\partial t}(u_x)^2\,\mathrm{d}x\right) \\ &= -\frac{T}{2}\int_{-\infty}^{\infty} \frac{\partial}{\partial t}(u_x)^2\,\mathrm{d}x \end{aligned}$$

We can take the derivative out of the integral because the integral is not with respect to t,

$$\frac{\mathrm{d}}{\mathrm{d}t}KE = -\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{T}{2}\int_{-\infty}^{\infty} (u_x)^2\right)\mathrm{d}x$$

The right side of the equation is the potential energy, and so we get the familiar result

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}KE &= -\frac{\mathrm{d}}{\mathrm{d}t}PE \\ \frac{\mathrm{d}}{\mathrm{d}t}(KE + PE) &= 0 \end{aligned}$$

When wave traveling, kinetic energy and potential energy experience exchange, and there is no net loss.

We can also generalize PE in three dimensions to

$$PE = \frac{T}{2}\int_{-\infty}^{\infty} |\nabla u|^2 \,\mathrm{d}\hat{\mathbf{x}}$$

2.6 The Diffusion Equation

Let us imagine a motionless liquid filling a straight tube or pipe and a chemical substance, say a dye, which is diffusing through the liquid . Simple diffusion is characterized by the following law: The dye moves from regions of higher concentration to regions of lower concentration. The rate of motion is proportional to the concentration gradient. (Fick's law). Let $u(x,t)$ be the concentration of the dye at position x of the pipe at time t.

In the section of the pipe from x_0 to x_1, the mass of the dye is

$$M(t) = \int_{x_0}^{x_1} u(x,t)\,\mathrm{d}x\,, \text{ so } \frac{\mathrm{d}M}{\mathrm{d}t} = \int_{x_0}^{x_1} u_t(x,t)\,\mathrm{d}x$$

The mass in this section of pipe cannot change except by flowing in and our of its ends. By Fick's law,

$$\frac{\mathrm{d}M}{\mathrm{d}t} = \text{flow in} - \text{flow out} = -ku_x(x_1,t) - ku_x(x_0,t)$$

Then

$$\int_{x_0}^{x_1} u_t(x,t)\,\mathrm{d}x = ku_x(x_1,t) - ku_x(x_0,t)$$

Differentiate with respect to x,

$$\boxed{u_t = ku_{xx}}$$

In three dimensions we have

$$\iiint_D u_t \,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \iint_{\mathrm{bdy}D} k(\hat{\mathbf{n}}\cdot\nabla u)\,\mathrm{d}S$$

By the divergence theorem, we get the three-dimensional diffusion equation

$$\boxed{u_t = k(u_{xx} + u_{yy} + u_{zz}) = k\delta u}$$

If there is an external source f, and if the rate k of diffusion is variable, we get the more general inhomogeneous

$$u_t = \nabla\cdot(k\nabla u) + f(x,t)$$

2.7 The Maximum Principle

With the diffusion equation defined

$$\begin{cases} u_t = ku_{xx} \\ 0 \leq x \leq l \\ 0 \leq t \leq T \end{cases}$$

$$\boxed{\text{The solution will attain its maximum and minimum value on the boundary } t = 0,\ x = 0,\ x = l}$$

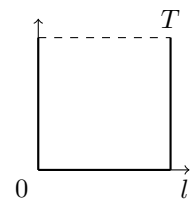

Figure 2.1: The solution will be at a maximum or a minimum along the bolded lines.

Proof: We will perturb the solution $u(x,t)$ with a small εx^2:

$$v(x,t) = u(x,t) + \varepsilon x^2$$

$$\begin{cases} v_t = u_t \\ v_{xx} = u_x x + 2\varepsilon \end{cases}$$

Then plug into the heat equation:

$$\begin{aligned} u_t - kv_{xx} &= u_t - k(u_{xx} + 2\varepsilon) \\ &= u_t - ku_{xx} - 2k\varepsilon \end{aligned}$$

The $u_t - ku_{xx}$ term should be zero, which means

$$v_t - kv_{xx} = -2k\varepsilon < 0 \tag{2.1}$$

We know this is negative because ε and k are positive. Suppose v attains its maximum value *inside* the boundaries. Then

$$\begin{cases} v_t = 0 \\ v_{xx} \leq 0 \end{cases}$$

Plugging these into (2.1):

$$0 \leq -kv_{xx} = -2k\varepsilon < 0$$

Where we know this is greater than or equal to zero because $v_x x$ is negative. The above is a contradiction: a number cannot be greater than or equal to zero and also less than zero. This means v only attains max along the boundary; it remains to show that this is also the case for u.

Recall $v = u + \varepsilon x^2$, so, by the definition of the maximum

$$v(x,t) \leq \max v(x,t) \text{ for } (x,t) \in \text{boundary}$$

And

$$\max v(x,t) \leq \max u + \varepsilon l^2 \text{ for } (x,t) \in \text{boundary}$$

,

so,

$$\begin{aligned} v(x,t) &\leq \max u + \varepsilon l^2 \text{ for } (x,t) \in \text{boundary} \\ u + \varepsilon x^2 &\leq \max u + \varepsilon l^2 \text{ for } (x,t) \in \text{boundary} \\ u &\leq \max u + \varepsilon(l^2 - x^2) \text{ for } (x,t) \in \text{boundary} \end{aligned}$$

Then let $\varepsilon \Rightarrow 0$, and

$$u \leq \max u \text{ for } (x,t) \in \text{boundary}$$

Which is precisely the definition of a maximum. WE can show the minimum is also along the boundary by proving this for $-u$ (the maximum of $-u$ is the minimum of u). We get

$$\min u \text{ for } (x,t) \in \text{boundary} \leq u \leq \max u \text{ for } (x,t) \in \text{boundary}$$

2.7.1 Applications

We can use the maximum principle to show **uniqueness**. For

$$\begin{cases} u_t - ku_{xx} = f(x,t) \\ u(x,0) = \phi(x) \\ u(0,t) = g(t) \\ u(l,t) = h(t) \end{cases}$$

If u_1 and u_2 are two solutions of above problem then $u_1 = u_2$. I.e., the above equation has a unique solution.

Proof: Consider $w = u_1 - u_2$. Then

$$\begin{cases} w_t - kw_{xx} = 0 \\ w(x,0) = 0 \\ w(0,t) = w(l,t) = 0 \end{cases}$$

The topmost equation is equal to zero because u_1 and u_2 both must satisfy the inhomogeneity, so, by linearity, the inhomogeneities are equal and their difference is zero. The boundary and initial conditions are equal to zero because of the same reason: they must satisfy the same equation.

Then, if the function is zero at the boundaries, and the maximum and minimum at the boundaries:

$$0 = \min w \leq w \leq \max w = 0$$

And thus $w = 0$ and $u_1 = u_2$.

Now we move on to **stability**. Our two equations are

$$\begin{cases} u_t = ku_x x \\ u(x,0) = \phi_1(x) \\ u(0,t) = u(l,t) = 0 \end{cases}$$
$$\begin{cases} u_t = ku_x x \\ u(x,0) = \phi_2(x) \\ u(0,t) = u(l,t) = 0 \end{cases}$$

Given these different initial conditions, how different will u_1, u_2 be?

Let $w = u_1 - u_2$, then

$$\begin{cases} w_t = kw_{xx} \\ w(x,0) = \phi_1 - \phi 2 \\ w(0,t) = w(l,t) = 0 \end{cases}$$

Then, because the boundaries are equal to zero,

$$w(x,t) \leq \max w \leq \max |\phi_1 - \phi_2| \text{ for } x \in [0,l]$$

And also

$$\min(\phi_1 - \phi_2) \leq w \leq \max(\phi_1 - \phi_2)$$

So that

$$|w| \leq \max |\phi_1 - \phi_2| \text{ for } x \in [0,l]$$

And thus

$$|u_1 - u_2| \leq \max |\phi_1 - \phi_2| \text{ for } x \in [0,l]$$

]

If we enforce small ε in $|\phi_1 - \phi_2| < \varepsilon$, u_1 and u_2 are then quite similar.

2.8 Solving the Heat/Diffusion Equation (Initial Condition)

Our equation is

$$\begin{cases} u_t = ku_{xx} \\ u(x,0) = \phi(x) \end{cases}$$

We proceed with the observations made in section 2.8.1.

Suppose $Q(x,t)$ is a solution. We want a particular solution. Use dilation invariance:

$$\frac{\sqrt{c}x}{\sqrt{ct}} \Rightarrow \frac{x}{\sqrt{t}}$$

Let $Q(x,t) = g(\frac{x}{\sqrt{t}})$, where g is arbitrary. For convenience, let $p = \frac{x}{\sqrt{t}}$. Then

$$\begin{aligned} Q_t &= g'\left(\frac{x}{\sqrt{t}}\right) \cdot -\frac{1}{2}\frac{x}{t^{3/2}} \\ Q_x &= g'\left(\frac{x}{\sqrt{t}}\right) \cdot \frac{1}{\sqrt{t}} \\ Q_{xx} &= g''\left(\frac{x}{\sqrt{x}}\right)\frac{1}{t} \end{aligned}$$

We know $Q_t - kQ_{xx} = 0$, so

$$\begin{aligned} -\frac{1}{2}\frac{x}{t^{3/2}}g' - k\frac{1}{t}g'' &= 0 \\ g'' + \frac{x}{k2\sqrt{t}}g' &= 0 \\ g'' + \frac{p}{2k}g' &= 0 \end{aligned}$$

Letting $y' = g''$, this is just a first order linear differential equation. Solving with an integrating factor, we eventually obtain

$$Q = g = C_1 \int_0^{x/\sqrt{t}} e^{-\frac{p^2}{4k}}\,\mathrm{d}p + C_2$$

Set Q to satisfy some initial condition, so that we may solve for its constants. For simplicity, we choose:

$$Q(x,0) = \begin{cases} 1 \text{ when } x > 0 \\ 0 \text{ when } x \leq 0 \end{cases}$$

Recall $t > 0$ by definition, and also by $x/\sqrt{t}$. Then, when $x > 0$, $x/\sqrt{t}$ approaches ∞ as $t \to 0$ (we can't plug in $t = 0$ directly):

$$C_1 \int_0^{\infty} e^{-\frac{p^2}{4k}}\,\mathrm{d}p + C_2 = 1$$

,

But when $x \leq 0$,

$$-C_1 \int_{-\infty}^{0} e^{-\frac{p^2}{4k}}\,\mathrm{d}p + C_2 = 0$$

Now we solve for C_1 and C_2. Adding the two equations above (the integrals cancel because the integrand is even):

$$2C_2 = 1 \Rightarrow C_2 = \frac{1}{2}$$

Now solve for C_1:

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-x^2} &= \sqrt{\pi} \\ \int_0^{\infty} e^{-\frac{p^2}{4k}}\,\mathrm{d}p &= \int_0^{\infty} e^{-u^2}\sqrt{4k}\,\mathrm{d}u \\ &= \sqrt{4k}\frac{\sqrt{\pi}}{2}\end{aligned}$$

And once again adding the two equations:

$$\sqrt{4k\pi}C_1 = 1 \Rightarrow C_1 = \frac{1}{\sqrt{4k\pi}}$$

And thus the particular solution is

$$Q(x,t) = \frac{1}{\sqrt{4k\pi}}\int_0^{x/\sqrt{t}} e^{\frac{p^2}{4k}}\,\mathrm{d}p + \frac{1}{2}$$

In section 2.8.1, we showed that the derivative of a solution is also a solution, as well as that we can shift the solution along the y axis, and convolve it, and the resulting function is still a solution. Now we put this to use. Let $Q_x = K$, and:

$$u(x,t) = \int_{\infty}^{\infty} K(x-y,t)\phi(y)\,\mathrm{d}y$$

Which is the convolution of Q and some initial condition ϕ. It can be shown that this u satisfies the initial condition but I omit the steps here.

Let us determine K:

$$\begin{aligned}\frac{\partial Q}{\partial x} &= \frac{\partial}{\partial x}\left(\frac{1}{\sqrt{4k\pi}}\int_0^{x/\sqrt{t}} e^{\frac{-p^2}{4k}}\,\mathrm{d}p + \frac{1}{2}\right) \\ &= \frac{\partial}{\partial x}\left(\frac{1}{\sqrt{\pi}}\int_0^{x/\sqrt{4kt}} e^{-u^2}\,\mathrm{d}u + \frac{1}{2}\right) \\ &= \frac{1}{\sqrt{4k\pi t}}e^{-(\sqrt{x}/2kt)^2}\end{aligned}$$

And we end up with

$$\boxed{K = \frac{1}{4k\pi t}\exp\left[-\left(\frac{x}{\sqrt{4kt}}\right)^2\right]}$$

Which is called kernel—it is effectively a Gaussian curve that spreads out and decreases as t increases. Then, the solution is

$$\boxed{u(x,t) = \frac{1}{\sqrt{4k\pi t}}\int_{\infty}^{\infty}\exp\left[-\left(\frac{|x-y|^2}{4kt}\right)\right]\phi(y)\,\mathrm{d}y}$$

Which is the heat kernel convolved with the initial condition.

Note: The heat kernel solution is only valid when the initial condition coincides with its Fourier series (see chapter 3). Thus, you might be tempted to use the heat kernel to show certain properties of the solution or of the heat equation in general. This is not recommended; instead, use the maximum principle of integration by parts for better generality.

2.8.1 Observations on the Solution

1. **Shift invariance:** If $u(x,t)$ is a sollution, $u(x-y,t)$ is still a solution.
2. **Divergence free:** If $u(x,t)$ is a solution, u_t, u_x, u_{xx} are still solutions. Note that this this means

$$\begin{aligned}(u_t)_t &= k(u_{xx})_t \\ (u_t)_t &= k(u_t)_{xx}\end{aligned}$$

   because we are free to change the order of derivatives.
3. If $u_1, u_2, \ldots, u_m$ are solutions, then their linear combinations are also a solution.
4. If $k(x,t)$ is a solution, then

$$u(x,t) = \int_{-\infty}^{\infty} k(x-y,t)g(y)\,\mathrm{d}y$$

   is also a solution, with arbitrary g. This is because

$$\begin{aligned}\frac{\partial}{\partial t}u(x,t) &= \frac{\partial}{\partial t}\int_{-\infty}^{\infty} k(x-y,t)g(y)\,\mathrm{d}y \\ &= \int_{-\infty}^{\infty} \frac{\partial}{\partial t}k(x-y,t)g(y)\,\mathrm{d}y\end{aligned}$$

   The same is true for the partial derivative with respect to x, allowing us to plug in to the heat equation and verify the equality. Note that $\int k(x-y,t)g(y)\,\mathrm{d}t$ is a **convolution** integral. In effect, this is the weighted sum of the values of k. g gives the weight.
5. **Dilation invariance:** If $u(x,t)$ is a solution, then $u(\sqrt{c}x, ct)$ is also a solution. By the chain rule:

$$cu(\sqrt{c}x, ct) = \sqrt{c}^2 u(\sqrt{c}x, ct)$$

2.9 Boundary Condition Solutions

2.9.1 Wave Equation with Dirichlet Condition

Our equation is

$$\begin{cases} u_{tt} = c^2 u_{xx} \\ u(0,t) = u(l,t) = 0 \end{cases}$$

We assume we are capable of separation of variables; that is, that the solution has the form

$$u(x,t) = X(x)T(t)$$

Now we plug back into the wave equation

$$\begin{aligned}
(X(x)T(t))_{tt} &= c^2(X(x)T(t))_{xx} \\
X(x)T''(t) &= x^2X''(t)T(t) \\
\frac{X''(x)}{X'(x)} &= \frac{T''(t)}{x^2T(t)} \\
-\frac{X''(x)}{X'(x)} &= -\frac{T''(t)}{x^2T(t)}
\end{aligned}$$

Where in the last line we added negative signs for convenience.

These will only be equal to each other if they are both equal to a constant:

$$-\frac{X''(x)}{X(x)} = -\frac{T''(t)}{T(t)} = \lambda$$

Now we have two ordinary differential equations:

$$\Big\{-X''(x) = \lambda X(x) - T''(t) = \lambda c^2 T(t)$$

Solve for X:

$$X = A\cos\sqrt{\lambda}x + B\sin\sqrt{\lambda}x$$

Note λ is positive, as we will prove later. For simplicity, let $\lambda = \beta^2$. Then

$$X = A\cos\beta x + B\sin\beta x$$

Now let us use the boundary condition, $u(0,t) = u(l,t) = 0$, to solve for some constants. We can set solely $X(x) = 0$, ignoring $T(t)$, because the whole expression is separable and we can divide both sides by $T(t)$:

$$\begin{cases} A = 0 \\ A\cos\beta l + B\sin\beta l = 0 \end{cases} \Rightarrow \begin{cases} A = 0 \\ B\sin\beta l = 0 \end{cases}$$

We don't want $B = 0$, because then the whole solution is zero, and this is a trivial solution. Then

$$\begin{aligned}
\sin\beta l &= 0 \\
\beta l &= n\pi, n \in \mathbb{Z} \\
\beta &= \frac{n\pi}{l}
\end{aligned}$$

Infinitely many β values to choose, hence

$$X(x) = B\sin\frac{n\pi}{l}x, n \in \mathbb{N}$$

Because $\sin -\theta = -\sin\theta$, we can restrict n to $1, 2, 3$, because you can put the negative sign in the constant.

Now let us solve the other ODE for T. We already know λ:

$$
\begin{aligned}
-\frac{T''(t)}{c^2T(t)} &= \left(\frac{n\pi}{l}\right)^2 \\
-T''(t) &= \left(\frac{cn\pi}{l}\right)^2 T(t) \\
T(t) &= C\cos\frac{cn\pi}{l}t + D\frac{cn\pi}{l}t
\end{aligned}
$$

Because there is n in the trigonometric functions, each constant is unique to the particular n:

$$
T_n(t) = C_n\cos\frac{cn\pi}{l}t + D_n\sin\frac{cn\pi}{l}t
$$

$u(x,t)$ is then

$$
\begin{aligned}
u(x,t) &= \sum_{n=1}^{\infty} X_nT_n \\
&= \sum_{n=1}^{\infty}\left(C_n\cos\frac{cn\pi}{l}t + D_n\sin\frac{cn\pi}{l}t\right)\sin\frac{n\pi}{l}x
\end{aligned}
$$

Then our solution for the boundary conditions is

$$
\boxed{u(x,t) = \sum_{n=1}^{\infty}\left(C_n\cos\frac{cn\pi}{l}t + D_n\sin\frac{cn\pi}{l}t\right)\sin\frac{n\pi}{l}x}
$$

Now we show this solution works with **initial conditions**:

$$
\begin{cases}
u_{tt} = c^2u_{xx} \\
u(0,t) = u(l,t) = 0 \\
u(x,0) = \phi(x) \\
u_t(x,0) = \phi(x)
\end{cases}
$$

We notice

$$
\phi(x) = \int_{n=1}^{\infty} C_n\sin\frac{n\pi}{l}x
$$

And

$$
\begin{aligned}
u_t(x,t) &= \sum_{n=1}^{\infty}\left(-C_n\frac{cn\pi}{l}\sin\frac{cn\pi}{l}t + D_n\frac{cn\pi}{l}\cos\frac{cn\pi}{l}t\right)\cdot\sin\frac{\pi t}{l}x \\
\psi(x) &= \sum_{n=1}^{\infty} D_n\frac{cn\pi}{l}\sin\frac{n\pi}{l}x
\end{aligned}
$$

Then separation of variables only works for functions that satisfy their sine Fourier series:

$$
\begin{cases}
\phi(x) = \sum_{n=1}^{\infty} C_n\sin\frac{n\pi}{l}x \\
\psi(x) = \sum_{n=1}^{\infty} D_n\frac{cn\pi}{l}\sin\frac{n\pi}{l}x
\end{cases}
$$

It can be shown that practically any odd function on the interval $(0,l)$ can be expressed in such a series.

2.9.2 Diffusion Equation with Dirichlet Condition

$$\begin{cases} u_t = ku_{xx}, 0 \leq x \leq l \\ u(0,t) = u(l,t) = 0 \end{cases}$$

We use the separation of variables method once again: assume

$$\begin{aligned} u(x,t) &= X(x)T(t) \\ X(x)T'(t) &= kX''(x)T(t) \\ -\frac{X''(x)}{X(x)} &= -\frac{T'(t)}{kT(t)} - \lambda \end{aligned}$$

X'' will be the same, but T' is a single derivative, so we must solve it differently (recall $\lambda = (n\pi/l)^2$):

$$\begin{aligned} X(x) &= B_n \sin \frac{n\pi}{l} x \\ -\frac{T'(t)}{kT(t)} &= \left(\frac{n\pi}{l}\right)^2 \\ T'(t) &= -k(\frac{n\pi}{l})^2 T(t) \\ T(t) &= C_1 \exp\left[-k\left(\frac{n\pi}{l}\right)^2 t\right], n = 1,2,3 \end{aligned}$$

And thus our solution is

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} C_n \exp\left[-k\left(\frac{n\pi}{l}\right)^2 t\right] \sin \frac{n\pi}{l} x}$$

And the initial value condition must satisfy

$$\phi(x) = u(x,0) = \sum_{n=1}^{\infty} c_n \sin \frac{n\pi}{l} x$$

2.9.3 Why Lambda Must be Positive

For the Dirichlet eigenvalue problem,

$$\begin{cases} -X'' = \lambda X \\ X(0) = X(l) = 0 \end{cases}$$

λ is always positive. **Proof:** If $\lambda = 0$,

$$\begin{aligned} X'' &= 0 \\ X(x) &= Cx + D \end{aligned}$$

Then, solving for constants:

$$\begin{cases} X(0) = D \\ X(l) = Cl + D \end{cases} \Rightarrow \begin{cases} D = 0 \\ C = 0 \end{cases}$$

Then $X(x) = 0$, and since we do not consider the zero vector an eigenvector, $\lambda = 0$ is not an eigenvalue. Now consider the case $\lambda < 0$. Taking λ negative such that $-\lambda$ is positive:

$$\begin{aligned} X'' &= (-\lambda)X \\ X'' &= \beta^2 X \\ X(x) &= Ce^{\beta x} + De^{-\beta x} \end{aligned}$$

Then, solving for the constants:

$$\begin{cases} X(0) = C + D \\ X(l) = Ce^{l\beta} = De^{-l\beta} \end{cases} \Rightarrow \begin{cases} C + D = 0 \\ Ce^{\beta l} + De^{-\beta l} = 0 \end{cases}$$

By the top equation, $C = -D$. We enforce $C \neq 0$ for the same reasons as above (it would result in the zero vector). Then:

$$\begin{aligned} Ce^{\beta l} - Ce^{\beta l} &= 0 \\ e^{\beta l} - e^{-\beta l} &= 0 \\ e^{\beta l} &= e^{-\beta l} \\ e^{2\beta l} &= 1 \\ \beta &= 0 \end{aligned}$$

But we enforced $\lambda < 0$! By contradiction, we have proved λ must be positive.

2.9.4 Wave Equation with Neumann Condition

$$\begin{cases} u_{tt} = c^2 u_{xx}, 0 \leq x \leq l \\ u_x(0,t) = u_x(l,t) = 0 \end{cases}$$

Consider

$$-X''(x) = \lambda X(x)$$

Because the Neumman condition has to do with the x derivative of u, we cannot make the same claims about λ. Then we consider three different cases. **Assume** $\lambda > 0$: Again, let $\lambda = \beta^2$

$$\begin{aligned} X(x) &= A\cos\beta x + B\sin\beta x \\ X'(x) &= -A\beta\sin\beta x + B\beta\cos\beta x \end{aligned}$$

Then, solving for the constants:

$$\begin{cases} X'(0) = B\beta \\ X'(l) = -A\beta\sin\beta l + B\beta\cos\beta l \end{cases} \Rightarrow \begin{cases} B\beta = 0 \Rightarrow B = 0 \\ -A\beta\sin\beta l = 0 \end{cases}$$

Then $\sin \beta l = 0$ and

$$\beta l = n\pi \Rightarrow \frac{n\pi}{l}, n = 1, 2, 3, \ldots$$

Then we have the exact same case as before, and we get the case $\lambda > 0$ implies

$$\begin{aligned} X_n(x) &= A_n \cos \frac{n\pi}{l} x \\ T_n(t) &= C_1 \cos \frac{cn\pi}{l} t + D_n \sin \frac{cn\pi}{l} t \end{aligned}$$

Then

$$u_n(x,t) = \left(C_n \cos \frac{2n\pi}{l} t + D_n \sin \frac{cn\pi}{l} \right) \cos \frac{n\pi}{l}$$

Now **assume** $\lambda = 0$, then:

$$\begin{aligned} X''(x) &= 0 \\ X(x) &= Cx + D \\ X'(x) &= C \end{aligned}$$

If we want

$$\begin{cases} X(0) = 0 \\ X(l) = 0 \end{cases}$$

we only need $c = 0$. I.e., $X(x) = D$, some constant. For T,

$$\begin{aligned} T''(t) &= 0 \\ T(t) &= At + B \end{aligned}$$

Then

$$\begin{aligned} u(x,t) = X(x)T(t) &= D(At + B) \\ &= At + B \end{aligned}$$

This is the particular solution for $\lambda = 0$, so denote

$$u_0(x,t) = A_0 t + B_0$$

At last **assume** $\lambda < 0$,

$$\begin{aligned} -X''(x) &= \lambda X(x) \\ X''(x) &= (-\lambda) X(x) \\ X''(x) &= \beta^2 X(x) \\ X(x) &= Ce^{\beta x} + De^{-\beta x} \end{aligned}$$

Then

$$X'(x) = C\beta e^{\beta x} - D\beta e^{-\beta x}$$

We attempt to solve for β and the constants:

$$\begin{cases} C\beta - D\beta = 0 \\ C\beta e^{\beta l} - D\beta e^{-\beta l} = 0 \end{cases} \Rightarrow \begin{cases} C = D \\ C\beta e^{\beta l} - C\beta e^{-\beta l} = 0 \end{cases}$$

Then, solving for β:

$$\begin{aligned} e^{\beta l} - e^{-\beta l} &= 0 \\ e^{2\beta l} &= 1 \\ \beta &= 0 \end{aligned}$$

WE have a contradiction again. We enforced $\lambda < 0 \Rightarrow \beta^2 > 0$. So, we may have $\lambda \geq 0$, but **not** $\lambda < 0$.

Then, the **general solution** to the Neumann condition is

$$\boxed{u(x,t) = A_0 t + B_0 + \sum_{n=1}^{\infty} \left(C_n \cos \frac{cn\pi}{l} t + D_n \sin \frac{cn\pi}{l} t \right) \cos \frac{n\pi}{l} x}$$

2.9.5 Diffusion Equation with Neumann Condition

$$\begin{cases} u_t = k u_{xx} \\ u_x(0,t) = u_x(l,t) = 0 \\ x(x,0) = \phi(x) \end{cases}$$

We take the same results as with the wave equation, except the time derivative is of order one. Starting with $\lambda = 0$:

$$\begin{aligned} X''(x) &= 0 \\ X(x) &= Cx + D \end{aligned}$$

But for the boundary condition to be satisfied, $C = 0$, and $X(x) = D$. And for T:

$$\begin{aligned} \frac{T'(t)}{kT(t)} &= 0 \\ T'(t) &= 0 \\ T_0(t) &= A_0 \end{aligned}$$

We combine these solutions (the constants combine into one) with the solutions for $\lambda > 0$:

$$u(x,t) = A_0 + \sum_{n=1}^{\infty} C_n \exp\left[-k \left(\frac{n\pi}{l} \right)^2 t \right] \sin \frac{n\pi}{l} x$$

Chapter 3

Fourier Series

Coming soon...